

Write a program to input 5 real numbers(floating point) and find their average. If the average is greater than 50 print good, otherwise print bad.

Write a program to check whether an int type number (taken as input) is a multiple of 5 or not.

Write a program to check whether an int type number (taken as input) is a 3-digit number divisible by three or not.

Write a program to accept a mark obtained by a student in computer science and print the grades accordingly:

<u>Marks</u>	<u>Grade</u>
Above 90	A
70 to 89	B
50 to 69	C
below 50	D

A cloth showroom has announced the following festival discounts on the purchase of items, based on the total cost of the items purchased:

Total Cost	Discount (in Percentage)
Less than ₹ 2000	5%
₹ 2001 to ₹ 5000	25%
₹ 5001 to ₹ 10000	35%
Above ₹ 10000	50%

Write a program to input the total cost and compute and display the amount to be paid by the customer after availing the discount.

An electricity company charges their consumers according to the units consumed per month according to the given traffic:

Units Consumed	Charges
Up to 100 units	₹ 2 per unit
More than 100 units and up to 200 units	₹ 1.80 per unit
More than 200 units	₹ 1.50 per unit

In addition to the above, every consumer has to pay ₹ 200 as Service Charge per month. Write a program to input the amount of units consumed and calculate the total charges payable (Bill) by the consumer.

A library charges a fine for books returned late. Following are the fines:

First five days : 40 paise per day.

Six to Ten days : 65 paise per day.

Above ten days : 80 paise per day.

Design a program to calculate the fine assuming that a book is returned N days late.

Using a switch statement, write a menu driven program to find the area and perimeter of a square or rectangle. For finding the area and perimeter of a square input the side length and for finding the area and perimeter of the rectangle input the length and breadth.

Using a switch statement, write a menu driven program to convert a given temperature from Fahrenheit to Celsius and vice versa. For an incorrect choice, an appropriate error message should be displayed.

(HINT : $C = \frac{5}{9} \times (F - 32)$ and $F = 1.8 \times C + 32$)

Write a menu driven program to find:

- Area of a rectangle, where length and breadth is taken as input.(Area=length *breadth)
- Area of a circle, where radius is taken as input.(Area= 22/7*radius²)

Write a menu driven program to input a month number between 1 to 12 and print the corresponding month name. That is for 1 display January, for 2 display February for 3 display March, etc.

Write a program to input three integers and check whether it forms a Pythagorean triplet or not. A set of three integers is said to be a Pythagoren triplet if the sum of the squares of any two integers is equal to square of the third integer. Example, (3, 4, 5), (5, 12, 13), and (7, 24, 25).

To foster a sense of water conservation, the water department has an annual water conservation tax policy. The tax are based on the water consumption of the consumer. The tax rates are as follows:

Water Consumed (in Gallons)	Tax Rate in ₹ /100 gallons
Upto 45	No Tax
More than 45 but 75 or less	475.00
More than 75 but 125 or less	750.00
More than 125 but 200 or less	1225.00
More than 200 but 350 or less	1650.00
More than 350	2000.00

Write a program to input the water consumed annually in gallons and output the tax payable.

Write a program to input the length of the 3 sides of a triangle (say a, b and c) and calculate the area depending upon the following criteria:

- If the sum of any two sides of the triangle is lesser than the third side then print "A triangle cannot be formed"
- If $a=b=c$ then print "It is an equilateral triangle" and calculate the area as

$$area = \frac{\sqrt{3}}{2} a^2$$

- If the sum of the squares of any two sides is equal to the square of the third side then print "It is a right-angled triangle" and calculate the area as

$$area = \frac{1}{2} * \text{product of the perpendicular sides}$$

- otherwise calculate area as $area = \sqrt{s(s-a)(s-b)(s-c)}$.

$$s = \frac{a+b+c}{2}$$
